

Game Design Concept - Secondary succession

1. Project overview

Working Title: Secondary succession

Genre: Farming, life simulation, exploration, ecosystem management

Perspective: Third person

Tone: Melancholic, hopeful, mysterious, cozy but strange

Player character: Ager, an alien botanist

Setting: A nameless, ruined planet

Primary progression: Discovery, technology, environment

1.1 High concept

Secondary succession is a sci-fi restoration farming and exploration game where an alien botanist lands on a ruined planet and uses alien technology to gradually rebuild functioning ecosystems.

1.2 Core fantasy

Be an alien botanist. Explore a ruined planet. Unearth planet's history. Learn proper conditions for each organism. Purify earth, air and water. Rebuild ecosystems. Watch the planet change.

2. Design pillars

2.1 Restoration over extraction

At the beginning of the game, the planet is totally barren and uninhabitable, with acid rains, radiation zones, frozen regions and contaminated soil. It becomes richer because of player's conscious actions: restoration of the atmosphere, regulation of the temperature, adjustment of the humidity, and cleansing of water and soil through lilies and fungi. Once areas become habitable, the player must carefully craft, balance and oversee ecosystems.

2.2 Every organism matters

Plants aren't isolated crops; they're components of living ecosystems. Overcrowding a habitat hinders the growth of individual plants. Clover adds nitrogen to soil, which

enables growth of nitrogen-sensitive plants. Fruit trees require pollen from other plants that flower at the same time.

2.3 Visible transformation

The player should constantly see evidence that the planet is recovering. Restoring the atmosphere changes the color of the sky, clouds form more often and dust storms become less frequent. Through cleansing the water, regular rain replaces acid rain, and soil returns to its natural color. Slowly insects begin appearing, and previously inaccessible areas become habitable. A new growth starts covering once-barren fields.

2.4 Discovery over instructions

Player learns ecological relationships through experimentation. Aeroponics provides an isolated environment for testing plants' preferences, like temperature, humidity, and required macronutrients. Bioscanner allows the player to track ecosystem's parameters, as well as individual organisms living in it. A failing ecosystem provides notes on what went wrong, allowing the player to correct their mistakes in the next attempt.

2.5 Hope and collapse

The ruined planet evokes feelings of melancholy. Ager will wonder what happened to it and if it could have been prevented. He might find an abandoned botanical laboratory, a seed bank, a decayed university greenhouse, or a bunker with the last agricultural and ecological reports – unsuccessful attempts of saving the planet by the previous culture. On the other hand, he will also explore run-down industrial zones, landfills, collapsed mines, destroyed power plants and abandoned data centers. But the dominant trajectory is renewal – learning from past mistakes and restoring the life that once were.

2.6 The planet is the farm

The ultimate goal of the game is restoring a living planet – the final ecosystem that is made up of smaller ecosystems. The player starts with their tiny aeroponics facility, and learns to purify small patches of land and water. Then they create their first small ecosystems, which slowly expand to larger areas. Ecosystems eventually cover the entire continent, at which point new continents are available for restoration – each with their own problems and unique species.

3. Core gameplay loop

3.1. Explore

Travel through a ruined planet searching for genetic samples, agricultural machinery, water sources, lost scientific records, and seed vaults. Discover what happened to the planet, find hints on what makes an ecosystems stable, recover seeds and technology.

3.2 Recover

Bring seeds, DNA samples, technology and biological specimens back to your base. Some are intact, some are damaged, some require experimentation to understand.

3.3 Cultivate

Use aeroponics to discover plants' requirements before exposing them to planet's hostile environment. Manage nutrients, light, temperature, and humidity to find the perfect conditions for each plant.

3.4 Experiment

Discover which plants can survive and transform environments – radiation, draught, cold, toxic soil. Crossbreed and adapt species.

3.5 Restore

Release mature plants into the world by crafting and balancing ecosystems. Deploy mechanical pollinators, humidity controllers, and radiation dampeners to keep them alive.

3.6 Observe

The ecosystems are changing. Some will become stable and transform their environment, while others will die out. Ecosystems can also interact with one another. Figure out what went wrong and try again.

3.7 Terraform

Once enough biological infrastructure exists, activate alien terraforming technology, changing temperature, atmospheric composition, humidity, water cycles and soil chemistry.

3.8 Explore further

The newly habitable environment opens previously inaccessible regions, allowing for further exploration.

4. Progression

4.1. Overview

The game's progression follows the gradual restoration of the planet. The player begins with a small aeroponic installation and a limited collection of seeds. Over time, they establish agricultural systems, restore damaged environments, reconstruct ecosystems, and develop increasingly advanced terraforming technology.

Each major region introduces a new environmental challenge while building upon systems and knowledge acquired previously. Progression gradually shifts from controlled cultivation toward ecosystem restoration, climate management, and eventually planetary self-sufficiency.

The world itself becomes the player's long-term project. Early regions require significant artificial support, while later restoration allows natural systems to take over an increasing share of environmental processes.

4.2. Restoration journey

Progression is primarily driven by the transformation of the world. A newly explored region may contain barren soil, abandoned infrastructure, contaminated water, or unstable environmental conditions. The player investigates the area, identifies its limitations, develops an appropriate restoration strategy, and introduces plants capable of rebuilding ecological processes.

As restoration progresses, the environment changes visibly. Plants spread beyond cultivated areas, soil becomes more fertile, water returns to damaged landscapes, and new organisms begin to appear. Restored ecosystems begin supporting themselves and influencing surrounding regions.

The core progression cycle is:

Discover > Cultivate > Restore > Expand > Stabilize

The final stages introduce another layer to this: the player returns to previously restored regions with new species and knowledge, completing ecological relationships that were impossible to establish earlier.

4.3. Regional progression

Each continent represents a major chapter in the restoration of the planet. Continents differ in geography, climate, available species, remnants of the previous culture, and the ecological problems the player must solve. The challenges become increasingly interconnected as the game progresses.

4.3.1 Region 1

This region is based on Europe. It contains degraded land, abandoned scientific settlements and damaged ecosystems. The primary challenge here is restoring the soil. However, the conditions are comparatively favorable, allowing the player to learn the fundamentals.

Key activities:

- Scavenging for seeds, technology, and lore
- Establishing basic aeroponic agriculture
- Analyzing seeds and learning optimal conditions
- Restoring soil fertility through plants
- Creating first ecosystems
- Learning how to stabilize ecosystem

When the player recovers this region, another one becomes unlocked.

4.3.2 Region 2

This region is based on North America. Rivers, lakes, wetlands, and groundwater systems have been disrupted, leaving large areas either dry or vulnerable to flooding.

Key activities:

- Reconnecting rivers and lakes
- Managing groundwater
- Restoring riverbanks
- Establishing aquatic ecosystems

Restoring a major river system and reconnecting it with the ocean unlocks another region.

4.3.3 Region 3

This region is based on Asia. Large areas contain industrial ruins, contaminated soil, and polluted water. The player begins using biological restoration methods to address environmental damage. Different organisms have specialized ecological and restoration properties. Some remove contaminants, others stabilize the soil or support the return of additional species. The first real bee appears, making artificial pollinators obsolete.

Key activities:

- Identifying different types and sources of contamination
- Cultivating resistant organisms
- Cleaning contaminated soil and water

- Recovering useful technology from abandoned infrastructure (filters)
- Transitioning artificially cleaned environments toward biological maintenance

Restoring all damaged environments through ecosystems opens up another region.

4.3.4 Region 4

This region is based on South America. By this stage, large areas of vegetation have returned, but many ecosystems remain incomplete and depend on each other. The player must reconstruct the relationships that allow complex ecosystems to function.

Key activities:

- Deploying pollinators
- Establishing herbivore populations
- Restoring predators
- Introducing decomposers and fungi
- Supporting seed dispersal
- Managing food chains
- Monitoring population balance

Establishing a stable ecosystem with balancing population unlocks the next region.

4.3.5 Region 5

This region is based on Africa. It introduces multiple environments where ecological restoration is heavily constrained by climate. Large areas experience extreme heat, drought, irregular rainfall and unstable seasonal conditions. The player gains access to advanced terraforming systems capable of influencing regional environmental conditions. Terraforming becomes increasingly interconnected with biological systems. Changes to climate can help one ecosystem while creating new problems elsewhere. The player is also encouraged to return to previous continents and make their ecosystems less dependent on technology.

Key activities:

- Managing regional temperature
- Influencing rainfall and humidity
- Restoring seasonal water cycles
- Protecting ecosystems from extreme conditions
- Coordinating large-scale environmental projects

Establishing stable regional climates capable of supporting large, diverse ecosystems unlocks the next region.

4.3.6 Region 6

This region is based on Australia. It presents an extreme combination of aridity, heat, fragile soils, water scarcity, and large distances between viable ecosystems. The player must combine previously developed systems to create persistent ecological networks in environments where natural recovery is extremely difficult.

Key activities:

- Establishing vegetation in arid regions
- Controlling salinity and land degradation
- Managing scarce water resources
- Creating connected habitats
- Protecting ecosystems from extreme conditions

Restoring this region opens up the final one.

4.3.7 Region 7

This region is based on Arctic, culminating in the player's journey to Svalbard and the Global Seed Vault (GSV). This region's ecosystems are already in perfect self-sustaining balance. The player reaches GSV and discovers multiple new plant varieties. These final species provide ecological functions that are missing from ecosystems established during the earlier stages of the game, allowing the player to turn off the terraformer.

Key activities:

- Exploring GSV
- Cultivating new species in controlled environment
- Identifying the ecological roles of the recovered species
- Returning to previously completed regions to complete missing ecological relationships
- Reducing dependence on terraforming infrastructure

When the player successfully balances all regions without relying on terraforming, the final challenge is unlocked.

4.3.8 The Planet

Each region can function independently, but the planet is not stable until the relationships between regions function as well. The player must observe how changes in one part of the planet influence distant ecosystems and identify the feedback loops that keep planetary systems in balance:

- Water cycle – vegetation releases moisture into the atmosphere, influencing rainfall and freshwater availability in other regions. Rivers, wetlands, forests, and oceans for a connected cycle.
- Carbon cycle – forests, grasslands, soils, wetlands and oceans absorb and release carbon at different rates. The player must establish ecosystems that maintain a stable planetary carbon balance.
- Migration – animals move between regions according to seasonal resources and environmental conditions. Bird migration can connect breeding grounds, feeding areas and winter habitats across multiple regions.
- Pollination and seed dispersal – plants and animals depend on one another across regional boundaries. Pollinators and seed-dispersing species allow vegetation to spread.
- Disturbance and recovery – fire, flood, drought, storms, and other natural disturbances are part of healthy ecosystem. Mature environments need to recover from these events without collapsing.

The final objective is to disable all remaining terraforming functions (balancing water, carbon, and preventing extreme conditions) and allow the planet to operate entirely through its restored natural systems. The player observes the planet through multiple seasonal cycles, identifies unstable feedback loops, and makes targeted changes to ecosystems where necessary. When the major planetary feedback loops become self-sustaining, the alien leaves the planet and game ends.